Towards Water-Sensitive Cities: Poster Presentation Script (v2)

Oct 28, 2026 · @Manaswi Raj

Verdict: keep the six-speaker model, change three things

Your structure works. One opener, one framework, three cases and one synthesis is the cleanest shape for 15 minutes and six people, and it maps onto the brief. v1 of the script is accurate and careful, but it would lose marks in three places.

1. **The four required dimensions are not visible.** The brief says every presentation must show case studies, technical explanation, political and social explanation, and solutions that are technocratic, bureaucratic and political-ecological. v1 analyses with three lenses but never labels its solutions that way. In v2 every case speaker follows the same four beats and names each one aloud, so the examiner can tick them off.
2. **Swap Melbourne for Delhi.** The brief's own example is the Delhi water crisis, and it asks for news, urban policy and specificity. Delhi has all three: Supreme Court hearings in June 2024, a 2025 CAG audit of the Delhi Jal Board, and a water crisis and a record flood in the same month. Melbourne has no readings in this course, so its political-ecology half is the thinnest. It survives as one line in the comparison. If another group takes Delhi, use Mumbai as the backup (Anand, Gandy and Parthasarathy are all readings).
3. **Use the course's vocabulary.** v1 reads like a policy brief. This is an urban political ecology class, so each case now carries its readings: Ranganathan on storm drains and tanker "mafias", Anand on pressure, Kaika and Swyngedouw on hidden networks, Roy on informality, Angelo on scale, Harvey on the right to the city, Hädrich Silva et al. on justice in water-sensitive design.

The three cases now each foreground one problem. Bengaluru is about **land**, Delhi about **distribution and fragmented authority**, and Zhengzhou about **technology at its limit under top-down rule**. Keep v1's evidence notes; they are good, and v2 extends them.

How the 15 minutes run

Every case speaker uses the same four beats, in the same order, and says the beat names out loud. Identical structure lets the listener compare cities without effort, and it proves the brief's four dimensions are covered.

1. **What happened:** one or two news events with dates and numbers (about 30 seconds).

2. **Technical explanation:** how water physically moved, and why the system failed (about 30 seconds).
3. **Political and social explanation:** who made the risk, who carried it, which reading explains it (about 45 seconds).
4. **Solutions in three registers:** one technocratic, one bureaucratic, one political-ecological (about 45 seconds).

Speaker	Part	Target time	Point at on the poster
1	Opening: the urban water paradox	1:30	Title band and the Delhi June 2024 contrast
2	Framework: three registers, five questions	2:15	Framework panel
3	Bengaluru: when land swallows lakes	2:30	Case column 1
4	Delhi: one month, two crises	2:30	Case column 2
5	Zhengzhou: a sponge is not a shield	2:30	Case column 3
6	Comparison, solutions, conclusion	2:15	Comparison table and commitments

That is 13:30 of speaking, leaving about 90 seconds for handovers and pointing. Spoken pace of about 140 words a minute means each case script is about 350 words. If you run long, cut examples inside a beat, never a whole beat.

Poster layout that matches the talk. A title band across the top with the one-line argument. A narrow framework panel on the left with the three registers and five questions. Three equal case columns in the middle, each with the same four rows (What happened, Technical, Political and social, Solutions), so the rows line up across cities. A comparison table and the five commitments at the bottom right, with references beneath. Use at most one headline number per case, with its source printed beside it.

Speaker 1: Opening, the urban water paradox (1:30)

Poster cue: the title band, then the two Delhi dates.

How can a city run out of water and drown in the same month?

(Pause.)

In June 2024, Delhi did exactly that. In the first weeks, during one of its worst heatwaves, people climbed onto water tankers to fill buckets. The Delhi government went to the Supreme Court asking for water from Himachal Pradesh. The court asked a sharper question back: if water is already arriving, where is it going?

Then, on 28 June, 228 millimetres of rain fell in 24 hours. That was the most for a June day since 1936. Roads and underpasses flooded, and part of a canopy at the airport collapsed.

Too much water and too little water look like opposite problems. We will argue they are one problem: how a city has rebuilt its land, rivers and aquifers, and for whom.

That is what the Anthropocene means at city scale. Humans are now a force reshaping the water cycle itself. We pave catchments, fill lakes, pump aquifers and change the climate that delivers the rain.

The water-sensitive city is the planners' answer: design the city to work with the water cycle instead of against it. Our argument is that this answer is necessary but not sufficient. A city becomes water-sensitive only when its technology, its institutions and its distribution of risk change together.

We test that claim in three cities: Bengaluru, Delhi and Zhengzhou.

Handover: "First, the framework we use to read each city: not just what the technology does, but who decides, who pays, and who carries the risk."

Speaker 2: Framework, three registers and five questions (2:15)

Poster cue: the framework panel.

A water-sensitive city treats rain, rivers, groundwater and wastewater as one connected cycle. The idea comes from Australian research by Wong and Brown in 2009. They describe cities moving through stages: first supplying water, then sewering it, then draining it away, and finally designing the whole city around the water cycle. China's Sponge City programme and India's push for rainwater harvesting belong to the same family. The shared idea is simple: rain is a resource to hold, not waste to throw away.

Urban political ecology adds one claim. Water in a city is never just nature. Neil Smith calls it produced nature: labour and capital have already reshaped it. Kaika and Swyngedouw show that once pipes go underground, the social relations behind them disappear from view. Water seems to arrive by itself, and nobody asks who was left unconnected.

So we read every case in three registers, and propose solutions in the same three.

- **Technocratic:** how does water physically move, and what is the design capacity? This matters, because water follows physics. But Ranganathan warns that cities fixate on technical fixes and hide the causes of risk.

- **Bureaucratic:** which agencies are responsible, who funds and maintains, and do departments coordinate? Ananya Roy adds that the state decides when its own rules apply. Informality is a mode of governing, not a gap in it.
- **Political-ecological:** who benefits, who is exposed and who is heard? Hädrich Silva and colleagues show that water-sensitive design can deepen inequality unless benefits and burdens are planned explicitly.

To keep this concrete, we ask five questions of every city: **Who decides? Who pays? Who benefits? Who carries the risk? And what does the water do?**

The last question matters. Political ecology does not say technology is irrelevant. It says technology never works in a vacuum.

Handover: "We start in Bengaluru, where the story begins with land."

Speaker 3: Bengaluru, when land swallows lakes (2:30)

Poster cue: case column 1; trace rain, lake, channel and borewell with your hand.

What happened. In September 2022, 131 millimetres of rain fell on Bengaluru in one night, and the Outer Ring Road, home to the city's tech campuses, went under water. Eighteen months later, in March 2024, the Chief Minister said 6,900 of the city's 14,000 borewells had dried up, and the city was short of about 500 of the 2,600 million litres it needed every day.

Technical explanation. Bengaluru was built on a cascade of about a thousand tanks, or keres, linked by channels. Each lake overflowed into the next, storing monsoon water and recharging the ground. Researchers at IISc counted 207 water bodies in Greater Bangalore in 1973 and 93 in 2010. WELL Labs' 2024 water balance shows the result: all six groundwater assessment units in urban Bengaluru are over-exploited, and natural recharge is only about a tenth of what the city uses each day. With channels severed, overflow now floods streets instead of filling the next lake.

Political and social explanation. Malini Ranganathan shows this was not an accident. After the 1898 plague, colonial planners recast the channels as sanitary drains. After the 1990s IT boom, wetlands became real estate. And when floods come, the state blames "encroachment", but enforces it selectively. Lower-middle-class settlers in Madina Nagar, on filled storm channels, face demolition. A multinational's offices built on wetland next door do not. Water access is just as uneven. In Ranganathan's fieldwork, only 10 to 30 percent of households on the unauthorised periphery had piped water; the rest relied on borewells and tanker operators who work alongside officials.

Solutions.

- **Technocratic:** reconnect lake cascades and protect valley zones; fill lakes with treated wastewater, which WELL Labs calls a large untapped resource; build recharge wells.
- **Bureaucratic:** put lakes, storm drains and sewage under one catchment-level authority, because they are split across municipal, water-board, planning and lake agencies today.
- **Political-ecological:** apply encroachment rules to corporate and state projects first, secure tenure before any eviction, and give lake users a seat in restoration decisions.

Handover: "Bengaluru shows how land decides water. Delhi shows what happens when no single authority controls it at all."

Speaker 4: Delhi, one month, two crises (2:30)

Poster cue: case column 2; point to the June 2024 timeline, then the basement photo or icon.

What happened. We opened with Delhi's June 2024. On 6 June the Supreme Court ordered Himachal Pradesh to release 137 cusecs of water to Delhi. Six days later the judges asked the Delhi government what it had done about tanker mafias, saying pipelines were running dry while tankers got water. Three weeks later the record rain arrived. And on 27 July 2024, three civil-services aspirants drowned when rain flooded the basement library of a coaching centre in Old Rajinder Nagar.

Technical explanation. Delhi depends on neighbouring states for about 90 percent of its raw water. Its official demand is about 1,290 million gallons a day, against supply of about 1,000. And of the water it treats, the Delhi Economic Survey 2023–24 classes 58 percent as non-revenue: lost to leaks, theft or unbilled use. On the flood side, drains were clogged and covered, so rain in Rajinder Nagar had nowhere to go but down into the basement.

Political and social explanation. Delhi's water crisis is made far beyond the city. Hillary Angelo warns against "methodological cityism", and here the crisis runs through Haryana's canals, Himachal's dams and fights between the Delhi government, the Lieutenant Governor and neighbouring states. Inside the city, water flows by pressure, as Nikhil Anand found in Mumbai: through councillors, tankers and contacts. A 2025 CAG audit found supply unequal across zones, and that a Delhi water policy begun in 2011 was still not framed. After Rajinder Nagar, officials blamed street-side "encroachers" for covering drains, and bulldozers removed small structures. Yet the illegal basement library, run by a profitable coaching industry, was the deadlier violation. This is Ranganathan's encroachment politics again.

Solutions.

- **Technocratic:** cut non-revenue water with metered districts and pipe replacement; recharge aquifers on the Yamuna floodplain; reuse treated wastewater; map and desilt every drain before the monsoon.
- **Bureaucratic:** finally adopt a Delhi water policy; give drains one owner, since they are split across several civic and state agencies; publish tanker GPS data and zone-wise supply.
- **Political-ecological:** equalise supply between rich and poor zones; recognise unauthorised colonies' claim to piped water; enforce building rules on powerful violators, not only vendors.

Handover: "Delhi's problem is fragmentation. Zhengzhou is the opposite: a powerful state, a national programme, and still a disaster."

Speaker 5: Zhengzhou, a sponge is not a shield (2:30)

Poster cue: case column 3; the sponge-city diagram, then the 201.9 mm figure.

What happened. In 2015 China's State Council set a national target: by 2030, 80 percent of urban built-up land should absorb and reuse 70 percent of its rainfall. Thirty pilot cities followed. Zhengzhou invested heavily. Then, on 20 July 2021, 201.9 millimetres of rain fell on the city in a single hour. The official investigation counted 380 people dead or missing in Zhengzhou, including passengers trapped on a metro train and drivers in a road tunnel.

(Pause after the figure. Do not dramatise the deaths.)

Technical explanation. Sponge measures, such as permeable paving, rain gardens and wetlands, work well for frequent, moderate rain. Kumar and colleagues, in our reading comparing India with China, note that their performance falls as storms grow. Over three days Zhengzhou received 617 millimetres, close to its average for a whole year. No sponge holds that. A city also needs pipes, pumps and safe overflow routes.

Political and social explanation. The State Council's own report did not blame only the rain. It found warnings were ignored and responses were slow, and that 139 of the dead and missing had been concealed. Investigative reporting on the report found that more than half of Zhengzhou's sponge spending, over 19 billion yuan, went on landscaping rather than drainage. That fits the course: Kaika and Swyngedouw show infrastructure turned into spectacle, and here visible greenery counted for more than buried pipes. Bulkeley's Anthropocene city appears too: the city as climate hero, measured by targets met rather than lives protected.

Solutions.

- **Technocratic:** pair green with grey, design for overflow beyond capacity, and floodproof underground transport.

- **Bureaucratic:** link each warning level to automatic action, such as suspending the metro on a red alert, and audit sponge spending independently.
- **Political-ecological:** publish casualty and risk data, protect a free press, and let residents of exposed areas shape where protection goes.

Handover: "Three cities, three failures of different kinds. Our last speaker brings them together."

Speaker 6: Comparison, solutions and conclusion (2:15)

Poster cue: the comparison table, then the five commitments.

City	Technical cause	What decided the outcome	Who carried the risk
Bengaluru	Lake cascade broken, aquifers over-pumped	Land markets, selective encroachment rules	Settlers on filled channels, tanker-fed periphery
Delhi	Imported water, high losses, blocked drains	Split authority, tanker economy, interstate politics	Poorer zones, unauthorised colonies, basement tenants
Zhengzhou	Rain far beyond any sponge design	Spending priorities, ignored warnings, concealment	Metro and tunnel users, exposed districts

Look down the middle column. In every city the technical means existed in some form. What decided the outcome was land, authority and accountability. Now look at the last column. Risk landed on the people with the weakest claim to land and services.

These cities differ in climate, size and politics, so we compare mechanisms, not magnitudes. And our argument needs two corrections. First, hydrology sets hard limits: no design absorbs 201.9 millimetres in an hour. Second, strong authority can help. Melbourne halved household water use during its Millennium Drought, to about 155 litres a person a day by 2010, because one regional manager could make every agency act together. Authority works when it answers to the public, which is what Zhengzhou lacked.

So what would a just water-sensitive city commit to?

1. **Plan by watershed, not by plot.** Protect lakes, channels and recharge zones before land is sold.
2. **Combine green and grey.** Sponges for everyday rain, pipes, pumps and overflow routes for the extreme.

3. **Count informal systems as infrastructure.** Tankers, settlers' networks and informal workers are part of how water moves.
4. **Enforce rules equally, and secure tenure first.** No flood park or drain clearance that evicts only the weak.
5. **Publish the data.** Supply by zone, warnings, budgets and losses, so failures can be corrected.

David Harvey calls the right to the city the right to change ourselves by changing the city. For water, that means a right to decide where water goes and whom it reaches. In the Anthropocene, the question is not only whether we can control water. It is whether we can live with it wisely, and fairly. Thank you.

Readings, Q&A prep and sources

Each speaker should be able to name their reading and its idea in one sentence if asked.

Reading	Speaker	The idea you use
Wong and Brown (2009); Brown, Keath and Wong (2009)	2	Stages toward the water-sensitive city
Smith, *The Production of Nature*	2	Urban water is produced nature
Kaika and Swyngedouw (2000)	2, 5	Buried networks hide social relations; infrastructure as spectacle
Roy (2005), *Urban Informality*	2, 4	The state decides when rules apply
Ranganathan (2015), *Storm Drains as Assemblages*	3, 4	Flood risk is produced; encroachment is enforced selectively
Ranganathan (2014), *Mafias in the Waterscape*	3, 4	Tanker operators as everyday public authority
Anand (2011), *Pressure*	4	Water flows by hydraulic and political pressure
Angelo, *Methodological Cityism*	4	The city's crisis is made beyond its boundary
Kumar et al. (2021), India vs China's Sponge City	5	Sponge measures weaken as storms grow
Bulkeley (2021), *Climate Changed Urban Futures*	1, 5	The Anthropocene city as climate hero

Reading	Speaker	The idea you use
Hädrich Silva et al. (2024)	2, 6	Water-sensitive design must plan who gains and who loses
Harvey (2008), *The Right to the City*	6	Collective control over how the city is remade

Likely questions, with short answers.

- *"Is the Anthropocene official?"* No. In March 2024 geologists rejected it as a formal epoch, but called it a valuable description of human impact. We use it that way.
- *"Isn't this just bad governance, not political ecology?"* Governance failures are not random. They fall on people with weak land claims, which is the political-ecology point.
- *"Does Zhengzhou prove sponge cities fail?"* No. They work within design limits and need grey infrastructure and accountable institutions beside them.
- *"Why not blame encroachers?"* We do blame encroachment, of every kind. The point is that enforcement picks the weakest encroacher.
- *"What would you do first in Delhi?"* Publish zone-wise supply and tanker data. Equal distribution cannot be argued for without it.

Accuracy notes.

- Zhengzhou's 380 dead or missing, 139 concealed and 201.9 mm in one hour come from the State Council investigation. The 19-billion-yuan landscaping finding is Sixth Tone's reporting of that report; attribute it that way.
- Bengaluru's 6,900 borewells and 500 MLD shortfall are the Chief Minister's statement of March 2024, not an audited count.
- Delhi's demand and supply figures vary by year and source (1,260–1,290 MGD demand; about 935–1,000 MGD supply). Say "about".
- Melbourne's 155 litres is a 2010 benchmark, not current use.
- Never say urbanisation alone causes every flood or dry borewell; say it changes how water moves.

Sources.

- Supreme Court on tanker mafias, June 2024 (Bar and Bench via Moneylife)
- Delhi's 228.1 mm on 28 June 2024 (Scroll, IMD data)
- Old Rajinder Nagar basement deaths and drain encroachment (Deccan Herald, PTI)
- CAG performance audit of Delhi Jal Board (2025), chapter 2
- Delhi Economic Survey 2022–23, water supply chapter
- Bengaluru 500 MLD shortfall and dry borewells (Deccan Herald)

- WELL Labs Bengaluru water balance, summarised (Newslaundry)
- IISc wetlands of Bangalore (Ramachandra and Aithal)
- Zhengzhou investigation findings (Sixth Tone)
- State Council sponge city guideline, 2015
- Course readings as listed in the table above.